Probability: Basic Concepts

Revision Notes for Tutorial and Problem-Solving Sessions

Name: Bittu Kumar **Date:** 22 August 2026

1. Basic Set Terminology

A **set** is a well-defined collection of objects. If x is an element of A, we write $x \in A$; otherwise, $x \notin A$.

$$A \subseteq B \iff x \in A \implies x \in B.$$

Two sets are equal precisely when each is contained in the other:

$$\boxed{A = B \iff A \subseteq B \text{ and } B \subseteq A.}$$

Important sets are:

$$\varnothing = \text{empty set}, \qquad \Omega = \text{universal set}.$$

The empty set is a subset of every set:

$$\varnothing \subseteq A \qquad \text{for every set } A.$$

2. Operations on Sets

Let $A, B \subseteq \Omega$.

$$\begin{aligned} \textbf{Union} \quad & A \cup B = \{x : x \in A \text{ or } x \in B\}, \\ \textbf{Intersection} \quad & A \cap B = \{x : x \in A \text{ and } x \in B\}, \\ \textbf{Complement} \quad & A^c = \Omega \setminus A, \\ \textbf{Difference} \quad & A \setminus B = \{x : x \in A,\ x \notin B\} = A \cap B^c. \end{aligned}$$

The sets A and B are called **disjoint** if

$$\boxed{A \cap B = \varnothing.}$$

Useful translations are:

Statement	Set notation
At least one of A, B	$A \cup B$
Both A and B	$A \cap B$
Only A	$A \setminus B$
Exactly one of A, B	$(A \setminus B) \cup (B \setminus A)$
Neither A nor B	$(A \cup B)^c$
Not all of A_i	$(\bigcap_i A_i)^c$

3. Fundamental Set Identities

Commutative laws:

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$A \cup B = B \cup A, \qquad A \cap B = B \cap A.$$

Associative laws:

$$A \cup (B \cup C) = (A \cup B) \cup C,$$

$$A \cap (B \cap C) = (A \cap B) \cap C.$$

Distributive laws:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Complement laws:

$$A \cup A^c = \Omega, \qquad A \cap A^c = \varnothing, \qquad (A^c)^c = A.$$

De Morgan's laws:

$$\boxed{\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c}$$

and

$$\boxed{\left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c.}$$

To prove a set equality $X = Y$, use the two-inclusion method:

$$X \subseteq Y \qquad \text{and} \qquad Y \subseteq X.$$

4. Cartesian Product and Power Set

The Cartesian product of A and B is

$$A \times B = \{(x, y) : x \in A,\ y \in B\}.$$

More generally,

$$A_1 \times \cdots \times A_n = \{(x_1, \ldots, x_n) : x_i \in A_i\}.$$

For example, the sample space of three coin tosses can be written as

$$\Omega = \{H, T\}^3.$$

The **power set** of A, denoted by 2^A, is the set of all subsets of A:

$$2^A = \{B : B \subseteq A\}.$$

If A has n elements, then

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\boxed{|2^A| = 2^n.}$$

5. Random Experiment, Sample Space, and Events

A **random experiment** is an experiment whose precise outcome is not known beforehand.
The **sample space** Ω is the set of all possible outcomes:

$$\omega \in \Omega.$$

An **event** is a subset of the sample space:

$$A \subseteq \Omega.$$

The event A occurs when the observed outcome satisfies

$$\omega \in A.$$

Examples:

Experiment	Sample space
One coin toss	$\{H, T\}$
Two coin tosses	$\{H, T\}^2$
Three coin tosses	$\{H, T\}^3$
Toss until first head	$\{H, TH, TTH, TTTH, \ldots\}$

An outcome is a single point ω, whereas an event is a collection of outcomes.

6. Sigma-Algebra

Let Ω be a sample space. A collection

$$\mathcal{F} \subseteq 2^\Omega$$

is called a **sigma-algebra** if:

$$\boxed{\begin{aligned} &\text{(i)} \quad \Omega \in \mathcal{F}, \\ &\text{(ii)} \quad A \in \mathcal{F} \implies A^c \in \mathcal{F}, \\ &\text{(iii)} \quad A_1, A_2, \ldots \in \mathcal{F} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}. \end{aligned}}$$

The elements of $\mathcal{F}$ are the admissible events.
From the definition:

$$\varnothing = \Omega^c \in \mathcal{F}.$$

Important examples are:

$$\{\varnothing, \Omega\}, \qquad 2^\Omega,$$

and, for $A \subseteq \Omega$,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\{\varnothing, A, A^c, \Omega\}.$$

The pair

$$(\Omega, \mathcal{F})$$

is called a **measurable space**.

7. Probability Measure and Probability Space

Let $(\Omega, \mathcal{F})$ be a measurable space. A probability measure is a function

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$$

satisfying the following axioms.

Axiom 1: Non-negativity

$$\mathbb{P}(A) \geq 0 \qquad \text{for every } A \in \mathcal{F}.$$

Axiom 2: Normalization

$$\boxed{\mathbb{P}(\Omega) = 1.}$$

Axiom 3: Countable additivity

If $A_1, A_2, \ldots$ are pairwise disjoint, then

$$\boxed{\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).}$$

The triple

$$\boxed{(\Omega, \mathcal{F}, \mathbb{P})}$$

is called a **probability space**.

8. Basic Properties of a Probability Measure

For $A, B \in \mathcal{F}$:

1. Probability of the empty set

$$\boxed{\mathbb{P}(\varnothing) = 0.}$$

2. Complement rule

Since $\Omega = A \dot{\cup} A^c$,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

and therefore

$$\boxed{\mathbb{P}(A^c) = 1 - \mathbb{P}(A).}$$

3. Monotonicity

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

If $A \subseteq B$, then

$$\boxed{\mathbb{P}(A) \leq \mathbb{P}(B).}$$

Indeed,

$$B = A \dot{\cup} (B \setminus A).$$

4. Difference rule
If $A \subseteq B$, then

$$\boxed{\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A).}$$

In general,

$$\boxed{\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B).}$$

5. Addition rule

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).}$$

If $A \cap B = \varnothing$, then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

9. Finite Uniform Probability Model

Suppose that Ω is finite and every outcome is equally likely. Then

$$\mathbb{P}(\{\omega\}) = \frac{1}{|\Omega|} \qquad \text{for every } \omega \in \Omega.$$

Therefore, for every event $A \subseteq \Omega$,

$$\boxed{\mathbb{P}(A) = \frac{|A|}{|\Omega|}.}$$

Thus,

$$\mathbb{P}(A) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

This formula is valid only when all elementary outcomes are equally likely.

10. Inclusion-Exclusion Principle

For two events,

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).}$$

For three events,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

For n events,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{\varnothing \neq J \subseteq \{1,\ldots,n\}} (-1)^{|J|+1} \mathbb{P}\left(\bigcap_{j \in J} A_j\right).$$

11. Union Bound and Bonferroni Bound

For a finite or countable collection of events,

$$\mathbb{P}\left(\bigcup_{i \in I} A_i\right) \leq \sum_{i \in I} \mathbb{P}(A_i).$$

This is called the **union bound** or Boole's inequality. Equality holds when the events are pairwise disjoint.

The first Bonferroni lower bound is

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \geq \sum_{i=1}^{n} \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j).$$

The union bound ignores all overlaps, whereas the Bonferroni bound includes the first overlap correction.

12. Conditional Probability

Let $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. The conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning on B means that B becomes the new effective sample space.

For a finite uniform sample space,

$$\mathbb{P}(A \mid B) = \frac{|A \cap B|}{|B|}.$$

In general,

$$\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A).$$

The conditioning event must satisfy

$$\mathbb{P}(B) > 0.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

13. Conditional Probability as a Probability Measure

Fix $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, and define

$$\mathbb{P}_B(A) = \mathbb{P}(A \mid B).$$

Then $\mathbb{P}_B$ is itself a probability measure:

$$\mathbb{P}_B(\varnothing) = 0, \qquad \mathbb{P}_B(\Omega) = 1.$$

If $A_1, A_2, \ldots$ are pairwise disjoint, then

$$\begin{aligned}\mathbb{P}_B\left(\bigcup_i A_i\right) &= \frac{\mathbb{P}\left(\left(\bigcup_i A_i\right) \cap B\right)}{\mathbb{P}(B)} \\ &= \frac{\sum_i \mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} \\ &= \sum_i \mathbb{P}_B(A_i).\end{aligned}$$

Consequently, all ordinary probability identities remain valid after conditioning:

$$\boxed{\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B).}$$

Also,

$$\boxed{\begin{aligned}\mathbb{P}(A \cup C \mid B) = \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B) \\ - \mathbb{P}(A \cap C \mid B).\end{aligned}}$$

14. Multiplication Rule

From the definition of conditional probability,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Therefore,

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B).}$$

Similarly,

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A).}$$

Hence,

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \mathbb{P}(B)\mathbb{P}(A \mid B).}$$

For three events,

$$\boxed{\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B \mid A)\mathbb{P}(C \mid A \cap B).}$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

15. General Chain Rule

For events $A_1, \ldots, A_n$, assuming the relevant conditional probabilities are defined,

$$\boxed{\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathbb{P}(A_1) \prod_{i=2}^{n} \mathbb{P}\left(A_i \middle| \bigcap_{j=1}^{i-1} A_j\right).}$$

Expanded explicitly,

$$\begin{aligned}\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1) \\ \times \mathbb{P}(A_3 \mid A_1 \cap A_2) \\ \times \cdots \\ \times \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).\end{aligned}$$

The chain rule is particularly useful for sequential experiments and sampling without replacement.

16. Partition of the Sample Space

A collection $\{B_i : i \in I\}$ is a partition of Ω if

$$B_i \cap B_j = \varnothing \qquad \text{whenever } i \neq j,$$

and

$$\bigcup_{i \in I} B_i = \Omega.$$

Thus, exactly one of the events B_i occurs.
For every event A,

$$A = \bigcup_{i \in I} (A \cap B_i),$$

where the events $A \cap B_i$ are pairwise disjoint.

17. Law of Total Probability

Let $\{B_i : i \in I\}$ be a finite or countable partition of Ω, with

$$\mathbb{P}(B_i) > 0.$$

Because

$$A = \bigcup_{i \in I} (A \cap B_i)$$

is a disjoint union,

$$\mathbb{P}(A) = \sum_{i \in I} \mathbb{P}(A \cap B_i).$$

Using the multiplication rule,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\mathbb{P}(A \cap B_i) = \mathbb{P}(B_i)\mathbb{P}(A \mid B_i).$$

Therefore,

$$\boxed{\mathbb{P}(A) = \sum_{i \in I} \mathbb{P}(B_i)\mathbb{P}(A \mid B_i).}$$

The law of total probability computes the overall probability of A by considering all mutually exclusive ways in which A can occur.

18. Bayes' Rule

Suppose $\{B_i : i \in I\}$ is a partition of Ω, and $\mathbb{P}(A) > 0$. Then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j \cap A)}{\mathbb{P}(A)}.$$

Using the multiplication rule and the law of total probability,

$$\boxed{\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j)\mathbb{P}(A \mid B_j)}{\sum_{i \in I} \mathbb{P}(B_i)\mathbb{P}(A \mid B_i)}.}$$

The components are:

Quantity	Interpretation
$\mathbb{P}(B_j)$	Prior probability
$\mathbb{P}(A \mid B_j)$	Likelihood
$\mathbb{P}(A)$	Evidence
$\mathbb{P}(B_j \mid A)$	Posterior probability

Bayes' rule reverses the direction of conditioning:

$$\mathbb{P}(A \mid B) \quad \longrightarrow \quad \mathbb{P}(B \mid A).$$

19. Independence of Two Events

Events A and B are independent, written

$$A \perp\!\!\!\perp B,$$

if

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).}$$

If $\mathbb{P}(B) > 0$, then

$$A \perp\!\!\!\perp B \quad \Longleftrightarrow \quad \mathbb{P}(A \mid B) = \mathbb{P}(A).$$

If $\mathbb{P}(A) > 0$, then

$$A \perp\!\!\!\perp B \quad \Longleftrightarrow \quad \mathbb{P}(B \mid A) = \mathbb{P}(B).$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Thus, independence means that knowing one event has occurred does not change the probability of the other.

Independence is symmetric:

$$A \perp\!\!\!\perp B \quad \Longleftrightarrow \quad B \perp\!\!\!\perp A.$$

20. Pairwise and Mutual Independence

Events $A_1, \ldots, A_n$ are **pairwise independent** if

$$\boxed{\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \qquad \text{for every } i \neq j.}$$

They are **mutually independent** if, for every subset

$$J \subseteq \{1, \ldots, n\}, \qquad |J| \geq 2,$$

we have

$$\boxed{\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j).}$$

For three events, mutual independence requires all four conditions:

$$\begin{aligned}
\mathbb{P}(A \cap B) &= \mathbb{P}(A)\mathbb{P}(B), \\
\mathbb{P}(A \cap C) &= \mathbb{P}(A)\mathbb{P}(C), \\
\mathbb{P}(B \cap C) &= \mathbb{P}(B)\mathbb{P}(C), \\
\mathbb{P}(A \cap B \cap C) &= \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).
\end{aligned}$$

Mutual independence implies pairwise independence, but pairwise independence does not necessarily imply mutual independence.

21. Quick Problem-Solving Guide

When reading a probability problem, use the following sequence.

1. Identify the random experiment and sample space Ω.
2. Define the relevant events using set notation.
3. Translate the wording:

At least one	union
Both or all	intersection
None	complement of a union
Only one	set difference
Given that	conditional probability

4. If the sample space is finite and uniform, use

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

5. For several overlapping events, use inclusion-exclusion.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

6. For an "at least one" upper estimate, use the union bound.
7. For sequential events, use the multiplication or chain rule.
8. If the sample space is divided into cases, use the law of total probability.
9. To reverse conditioning, use Bayes' rule.
10. Use independence only after verifying

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

End of Concept Revision

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.